

Section 9.1: Introduction

Goals:

- to examine mathematical properties of point estimators
 -
- to derive point estimators
 -

Section 9.2: Relative Efficiency

Motivation: Often, there is more than one unbiased estimator for a target parameter, and we want to determine the *best* one.

Def. Given two unbiased estimators of θ , $\hat{\theta}_1$ and $\hat{\theta}_2$, the **efficiency** of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is

Question. What does relative efficiency tell us?

Example 1. Suppose $Y \sim N(\mu, \sigma)$, and we wish to estimate μ . Calculate the efficiency of the sample median relative to the sample mean.

Example 2. Let $Y_1, \dots, Y_n$ be a random sample from a Uniform distribution on the interval $(0, \theta)$. Let $\hat{\theta}_1 = 2Y_1$ and $\hat{\theta}_2 = 2\bar{Y}$.

(a) Show that $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators of θ .

(b) Calculate the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$.

Cramer-Rao Inequality. Suppose $Y_1, \dots, Y_n$ are i.i.d. with pdf $f(y)$ (which has θ in it). If $\hat{\theta}$ is an unbiased estimator of θ , then under very general conditions,

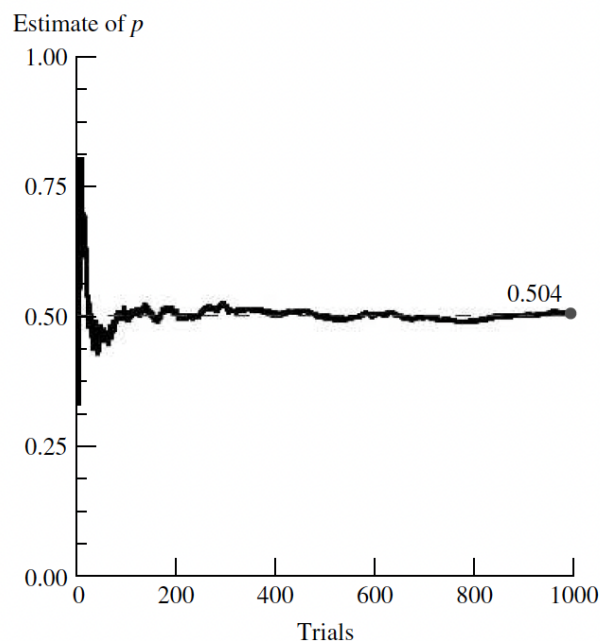
Notes:

- $I(\theta)$ is the _____.
 - If $V(\hat{\theta}) = 1/I(\theta)$, then $\hat{\theta}$ is _____.
- i.e.,

Example 3. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Exponential}(\theta)$. Is the sample mean an efficient estimator of θ ?

Section 9.3: Consistency

Motivation: Consider tossing a fair coin n times. We estimate $p = P(H)$ using $\hat{p} = Y/n$, where Y is the number of heads that occur. As n gets larger, $\hat{p}$ should get closer and closer to the true value of p .



In particular, the **error of estimation**, $|\hat{\theta} - \theta| =$ _____, should get smaller and smaller as $n \rightarrow \infty$.

Def. The estimator $\hat{\theta}_n$ is a **consistent estimator** of θ if,

or, equivalently,

Remark. Synonymously, we say _____.

Recall:

Tchebysheff's Theorem: Let Y be a RV with mean μ and variance σ^2 . Then, for any constant $k > 0$,

$$P(|Y - \mu| < k\sigma) \geq 1 - \frac{1}{k^2}.$$

Equivalently,

$$P(|Y - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

Theorem 9.1. If $\hat{\theta}_n$ is an unbiased estimator of θ and $\lim_{n \rightarrow \infty} V(\hat{\theta}_n) = 0$, then

Proof.

Law of Large Numbers. Let $Y_1, \dots, Y_n$ be an i.i.d. sample from a distribution with mean μ and variance $\sigma^2 < \infty$. Then,

Proof.

Example 4. Let $X_1, \dots, X_n$ be i.i.d. with $E(X_i) = \mu_1$ and $V(X_i) = \sigma_1^2 < \infty$. Let $Y_1, \dots, Y_n$ be i.i.d. with $E(Y_i) = \mu_2$ and $V(Y_i) = \sigma_2^2 < \infty$. Assume the X 's and Y 's are independent. Show that $(\bar{X}_n - \bar{Y}_n) \xrightarrow{p} (\mu_1 - \mu_2)$.

Theorem 9.2. Suppose that $\hat{\theta}_n \xrightarrow{p} \theta$ and $\hat{\gamma}_n \xrightarrow{p} \gamma$. Then,

(a) $(\hat{\theta}_n + \hat{\gamma}_n) \xrightarrow{p}$

(b) $(\hat{\theta}_n - \hat{\gamma}_n) \xrightarrow{p}$

(c) $(\hat{\theta}_n \cdot \hat{\gamma}_n) \xrightarrow{p}$

(d) If $\gamma \neq 0$, then $\frac{\hat{\theta}_n}{\hat{\gamma}_n} \xrightarrow{p}$

(e) If $g(\cdot)$ is a real-valued function that is continuous at θ , then

Example 5. Reconsider the previous example. Using Theorem 9.2, show that $(\bar{X}_n - \bar{Y}_n) \xrightarrow{p} (\mu_1 - \mu_2)$.

Example 6. Suppose $Y_1, \dots, Y_n$ are i.i.d. such that $E(Y_i) = \mu$, $V(Y_i) = \sigma^2$, $E(Y_i^2) = \mu'_2$, and $E(Y_i^4) = \mu'_4$ are all finite. Prove that $S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y}_n)^2 \xrightarrow{p} \sigma^2$.

Recall. For $Y_1, \dots, Y_n$ i.i.d. with $E(Y_i) = \mu$ and $V(Y_i) = \sigma^2$, we used the CLT to derive a $100(1 - \alpha)\%$ CI for μ :

For large n and σ unknown, we can replace σ with s and still have a $100(1 - \alpha)\%$ CI for μ .

But Why?

Theorem 9.3. Suppose U_n is a random variable whose distribution converges to $N(0, 1)$ as $n \rightarrow \infty$. If $W_n \xrightarrow{p} 1$, then

Section 9.4: Sufficiency

Motivation: We want to focus on the essential aspects of our data and ignore information that's not necessary to conduct inference.

Def. A **sufficient statistic** is

Example 7. Consider the outcomes of n trials of a binomial experiment, $X_1, \dots, X_n$, where

$$X_i = \begin{cases} 1, & \text{if trial } i \text{ is a success} \\ 0, & \text{if trial } i \text{ is a failure} \end{cases}.$$

For a constant probability of success, p , $Y = \sum_{i=1}^n X_i \sim \text{Bin}(n, p)$.

Question of interest: If we know the number of successes among the n trials (i.e., if we know the value of Y), can we gain more information about p by looking at other functions of $X_1, \dots, X_n$?

Answer:

Def. Let $Y_1, \dots, Y_n$ denote a random sample from a probability distribution with unknown parameter θ . Then, $U = g(Y_1, \dots, Y_n)$ is a **sufficient statistic** if

Otherwise, U is not a sufficient statistic.

Example 8. Let $Y = \sum_{i=1}^n X_i \sim \text{Bin}(n, p)$ with $n > 2$. Is $T = X_1 + X_2$ a sufficient statistic for p ?

Remark. The previous definition tells us how to *check* whether a statistic is sufficient, but it does not tell us how to *find* a sufficient statistic.

NEW Notation for Probability Functions:

- Discrete:
 - Univariate:
 - Multivariate:
- Continuous:
 - Univariate:
 - Multivariate:

Def. For random variables $Y_1, \dots, Y_n$ whose distribution depends on a parameter θ , the **likelihood function** is

Remark. The likelihood function simplifies further when independence is present.

- Discrete case: Let $Y_1, \dots, Y_n$ be an i.i.d. sample from a distribution with pmf $p(y|\theta)$. Then,
- Continuous case: Let $Y_1, \dots, Y_n$ be an i.i.d. sample from a distribution with pdf $f(y|\theta)$. Then,

Theorem 9.4: Factorization Criterion. Let U be a statistic based on the random sample $Y_1, \dots, Y_n$. Then U is a *sufficient statistic* for θ if and only if

where $g(\cdot)$ and $h(\cdot)$ are nonnegative functions, $g(u, \theta)$ is a function of only u and θ , and $h(y_1, \dots, y_n)$ is not a function of θ .

Example 9. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Exponential}(\theta)$. Using the Factorization Criterion, show that $\bar{Y}$ is a sufficient statistic for θ .

Example 10. Consider Example 7. Using the Factorization Criterion, show that $Y = \sum_{i=1}^n X_i$ is a sufficient statistic for p .

Example 11. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Rayleigh}(\theta)$. Find a sufficient statistic for θ .

Note: The Rayleigh distribution is often used by engineers to model the lifespan of electronics. The pdf of this distribution is

$$f(y|\theta) = \begin{cases} \frac{2y}{\theta} e^{-y^2/\theta}, & y > 0 \\ 0, & \text{o.w.} \end{cases}.$$

Example 12. Let $Y_1, Y_2, \dots, Y_n$ denote a random sample from a $U(0, \theta)$ distribution. Show that $Y_{(n)} = \max(Y_1, Y_2, \dots, Y_n)$ is sufficient for θ .

Facts:

- Any one-to-one function of a sufficient statistic is also sufficient.
- The data itself is **always** a sufficient statistic.

Section 9.5: The Rao-Blackwell Theorem and Minimum-Variance Unbiased Estimation

Theorem 9.5: Rao-Blackwell Theorem. Let $\hat{\theta}$ be an unbiased estimator for θ such that $V(\hat{\theta}) < \infty$. If U is a sufficient statistic for θ , define $\hat{\theta}^* = E(\hat{\theta}|U)$. Then,

Proof.

Def. A minimal sufficient statistic (MSS) is

Remark. A MSS is not unique—any one-to-one function of a MSS is also minimal sufficient.

Fact. An unbiased estimator that is a function of a MSS is a *minimum variance unbiased estimator (MVUE)*. In other words,

Lehmann-Scheffé Technique. Suppose $Y_1, \dots, Y_n$ have the likelihood function $L(y_1, \dots, y_n | \theta)$.

1. Calculate $R = \frac{L(x_1, \dots, x_n | \theta)}{L(y_1, \dots, y_n | \theta)}$.
2. Find a function $g(\cdot)$ such that R is free of θ if and only if $g(x_1, \dots, x_n) = g(y_1, \dots, y_n)$.
3. $U = g(Y_1, \dots, Y_n)$ is a MSS for θ .

Example 13. Let $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$.

(a) Find a MSS for p .

(b) Find a MVUE for p .

Example 14. Let $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Rayleigh}(\theta)$.

(a) Find a MSS for θ .

(b) Find a MVUE for θ .

Example 15. Let $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Find MVUEs for μ and σ^2 .

Example 16. Let $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Exponential}(\theta)$. Find a MVUE for $V(Y_i)$.

Section 9.6: The Method of Moments**Def.**

- $\mu'_k = E(Y^k)$ is the _____.
- $m'_k =$ _____ is the _____.

Method of Moments (MOM):

Choose as estimators the solutions to the equations $\mu'_k = m'_k$, for $k = 1, 2, \dots, t$, where t is the number of parameters to be estimated.

Example 17. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} U(0, \theta)$.

(a) Estimate θ using the method of moments.

(b) Show that your estimator is a consistent estimator of θ .

Fact. Sample moments are consistent estimators of the corresponding population moments; i.e.,

Example 18. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Gamma}(\alpha, \beta)$.

- (a) Estimate α and β using the method of moments.

(b) Show that $\hat{\alpha}$ and $\hat{\beta}$ are consistent estimators of α and β .

(c) Are $\hat{\alpha}$ and $\hat{\beta}$ MVUEs?

Remark. A MOM estimator is **not** necessarily a MVUE.

Section 9.7: The Method of Maximum Likelihood

So far, we have discussed the following estimation procedures:

1. Find a function of a MSS that is an unbiased estimator for the parameter of interest.
 - Problem:
2. Method of Moments
 - Problem:

Remark. The method of maximum likelihood is a direct procedure that usually results in MVUEs!

Method of Maximum Likelihood: Suppose that the likelihood function depends on k parameters $\theta_1, \theta_2, \dots, \theta_k$. Choose as estimates the values of the parameters that maximize the likelihood

Example 19. Let $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Bernoulli}(p)$. Find the maximum likelihood estimator (MLE) of p .

Example 20. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} \text{Exponential}(\theta)$. Find the MLE of θ .

Example 21. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Find the MLEs of μ and σ^2 .

Fact. If U is **any** sufficient statistic, then the MLE is **always** a function of U .

- Thus, the MLE is _____.
- So, if the MLE is unbiased, it will be a _____ (by a previous Fact).

Invariance Property of MLEs. If $t(\theta)$ is a one-to-one function of θ and $\hat{\theta}$ is the MLE for θ , then

Example 22. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Find the MLE of σ .